Contents

1 Budget Estimate

1.1 Detailed Budget Breakdown for 4-Year PhD Program

Comprehensive Financial Planning for AI-IDS Research
Last Updated: December 22, 2025
Total Estimated Budget: $53,000 over 4 years

1.2 Table of Contents

1.3 1. Budget Summary

Category	Amount	Percentage
Equipment & Hardware	$18,000	34%
Computational Resources (Cloud)	$15,000	28%
Conference Travel	$10,000	19%
Publications (Open Access)	$3,500	7%
Personnel (Undergrad Support)	$3,000	6%
Software Licenses	$2,500	5%
Miscellaneous	$1,000	2%
TOTAL	**$53,000**	**100%**

1.4 2. Personnel Costs

1.4.1 2.1 PhD Student Stipend

Amount: $0 (Covered by Research Assistantship)
Notes: - Typical PhD stipend: $25-35K/year - Covered by department/advisor funding - Not included in research budget

1.4.2 2.2 Undergraduate Research Assistant

Amount: $3,000
Breakdown: - Summer research support (10 weeks × $15/hour × 20 hours/week) = $3,000 - Help with data collection, testbed setup, documentation

Timeline: - Year 2 Summer: $1,500 (testbed setup) - Year 3 Summer: $1,500 (experiments and documentation)

Justification: Undergraduate assistance accelerates testbed deployment and documentation, freeing PhD student for advanced research tasks.

1.5 3. Equipment and Hardware

Total: $18,000

1.5.1 3.1 Network Testbed Equipment

Item	Quantity	Unit Price	Total	Justification
OpenFlow SDN Switches	5	$1,000	$5,000	Enable SDN experiments, OpenFlow protocol testing
High-Performance Servers	2	$4,000	$8,000	Fog nodes for federated learning aggregation
Raspberry Pi 4 (8GB)	10	$100	$1,000	Edge computing nodes for distributed IDS
IoT Devices (Mixed)	50	$50	$2,500	Smart sensors, cameras for IoT traffic generation
Network Cables & Accessories	Bulk	-	$500	CAT6 cables, patch panels, connectors
External Storage (Backup)	2	$500	$1,000	10TB NAS for dataset and model backup

Subtotal: $18,000

Purchase Timeline: - Year 1 Q4: External storage ($1,000) - Year 2 Q3: All testbed equipment ($17,000)

Justification: Physical testbed is essential for real-world validation of distributed IDS. Cannot be replaced by simulation alone. Equipment enables: - SDN attack scenarios (controller DoS, flow table overflow) - Edge-fog-cloud hierarchical deployment - IoT malware propagation studies - Latency and throughput measurements on real hardware

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1.6 4. Computational Resources (Cloud)

Total: $15,000

1.6.1 4.1 Cloud GPU Instances

Provider	Instance Type	Usage	Cost
AWS EC2	p3.8xlarge (4× V100)	5,000 hours	$12,000
Google Cloud	n1-standard-8 (CPU backup)	2,000 hours	$1,500
Storage	S3/Cloud Storage (5TB)	4 years	$1,500

Usage Breakdown: - Year 1: 1,000 hours (model prototyping) - $2,400 - Year 2: 1,500 hours (distributed training) - $3,600 - Year 3: 2,000 hours (extensive experiments) - $4,800 - Year 4: 500 hours (final experiments) - $1,200

Justification: - Deep learning models (CNN-LSTM, Transformer) require GPU acceleration - Large-scale experiments need parallel processing - University GPU cluster has limited availability and queue times - Cloud provides on-demand access for deadline-driven research

Cost Optimization: - Use spot instances (60-70% discount) - Apply for AWS/Google Cloud research credits ($3-5K potential) - Batch experiments during off-peak hours

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1.7 5. Software Licenses

Total: $2,500

Software	Annual Cost	Years	Total	Purpose
MATLAB (if needed)	$500	2	$1,000	Signal processing, network simulation
Network Analysis Tools	$750	2	$1,500	Commercial packet analysis, deep inspection
Total	-	-	**$2,500**	-

Notes: - Most tools are open-source (TensorFlow, PyTorch, Scikit-learn) - MATLAB may not be needed if Python alternatives suffice - University site licenses may cover some costs

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1.8 6. Conference Travel

Total: $10,000

1.8.1 6.1 Conference Breakdown

Year	Conference	Location	Estimated Cost	Purpose
Year 1	IEEE Workshop	Domestic	$1,500	First paper presentation
Year 2	IEEE INFOCOM	International	$3,000	Major networking conference
Year 3	USENIX Security	US/International	$3,000	Top security venue
Year 3	Local Workshop	Domestic	$500	Additional exposure
Year 4	IEEE S&P	US	$2,000	Top-tier security conference

Cost Breakdown per Conference: - Airfare: $400-800 (domestic), $1,000-1,500 (international) - Hotel: $150-250/night × 4 nights = $600-1,000 - Registration: $500-800 (student rate) - Meals: $50/day × 5 days = $250 - Ground transportation: $100-200

Funding Sources: - Conference travel grants: $500-1,000/conference - Department travel funds: $500-1,000/year - Advisor research grants: $1,000-2,000/year - Out-of-pocket: Remainder

Justification: - Conference presentations establish research credibility - Networking with leading researchers in field - Feedback improves research quality - Required for career development

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1.9 7. Publications

Total: $3,500

1.9.1 7.1 Open Access Fees

Publication Type	Cost per Paper	Number	Total
Conference Open Access	$400	2	$800
Journal Open Access	$1,500	2	$3,000
Total	-	**4**	**$3,800**

Breakdown: - IEEE Transactions (TIFS, TMC): $1,495-1,995 per paper - ACM journals: $1,500-2,000 per paper - Conference open access: $300-500 per paper - arXiv preprints: Free

Budget Allocation: - Year 2: $800 (1 conference open access) - Year 3: $1,500 (1 journal open access) - Year 4: $1,200 (1 journal, 1 conference)

Justification: - Open access increases citation count and impact - Required by some funding agencies (e.g., NSF, EU grants) - Broader dissemination to industry and practitioners

Cost Reduction: - Some journals offer waivers for students - Author's accepted manuscript on institutional repository (free) - Negotiate with publishers for reduced fees

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1.10 8. Miscellaneous

Total: $1,000

Item	Amount	Purpose
Books and Online Resources	$300	Textbooks, O'Reilly subscriptions
Printing and Materials	$200	Poster printing, thesis binding
Professional Memberships	$200	IEEE, ACM student membership
Contingency Fund	$300	Unexpected expenses

1.11 9. Budget Timeline

1.11.1 Year 1: $4,000

- Cloud GPU: $2,400
- Storage (backup): $1,000
- Books & resources: $200
- Professional memberships: $100
- Conference travel: $300 (local workshop)

1.11.2 Year 2: $17,900

- **Major spending year (testbed equipment)**
- Testbed hardware: $17,000
- Cloud GPU: $3,600
- Undergrad support: $1,500
- Software licenses: $1,000
- Conference travel: $3,000

1.11.3 Year 3: $18,300

- Cloud GPU: $4,800
- Undergrad support: $1,500
- Conference travel: $3,500
- Publications (OA): $1,500
- Software licenses: $1,000

1.11.4 Year 4: $12,800

- Cloud GPU: $1,200
- Conference travel: $2,000
- Publications (OA): $1,200
- Thesis printing: $200
- Miscellaneous: $300

Cumulative Total: $53,000

1.12 10. Funding Sources

1.12.1 10.1 Primary Funding

University Research Assistantship - Covers stipend and tuition - $25-35K/year (living expenses) - Health insurance

1.12.2 10.2 Research Funding

Source	Amount	Likelihood	Timeline
Department Research Grants	$5-10K	High	Year 1-2
Advisor's Research Grants	$10-20K	High	Year 2-4
External Fellowships	$10-30K	Medium	Year 2-3
Cloud Provider Credits	$3-5K	High	Year 1-2
Conference Travel Grants	$2-4K	Medium	Year 2-4
Industry Partnerships	$5-15K	Medium	Year 3-4

1.12.3 10.3 Fellowship Opportunities

NSF Graduate Research Fellowship Program (GRFP) - Award: $37K/year stipend + $12K tuition - Apply: Year 1 or 2 - Likelihood: Competitive (15-20% acceptance)

Industry Fellowships - Google PhD Fellowship: $50K/year - Facebook Fellowship: $42K/year - Microsoft Research Fellowship: $28K/year - Cisco Research Fellowship: $25K/year

1.12.4 10.4 Cloud Research Credits

- **AWS Educate:** $100-200 credits
- **AWS Research Credits:** $5,000-10,000 (application required)
- **Google Cloud for Research:** $5,000-20,000
- **Microsoft Azure for Research:** $10,000-20,000

1.13 11. Justification

1.13.1 11.1 Equipment Justification

Why Physical Testbed? ($18,000) - Simulation cannot replicate real-world latency, packet loss, hardware constraints - IoT device behavior differs significantly in simulation vs. reality - Required for latency validation (<100ms target) - Edge device resource constraints (CPU, memory) only testable on real hardware - Publications expect real-world validation for acceptance

Why Multiple Devices? - 5 SDN switches: Create realistic network topology with multiple paths - 10 Edge nodes: Test scalability and federated learning with realistic number of participants - 50 IoT devices: Generate realistic heterogeneous traffic patterns

1.13.2 11.2 Computational Resources Justification

Why Cloud GPU? ($15,000) - Deep learning models require 100-1000× speedup with GPUs - University GPU cluster: - Long queue times (days to weeks) - Limited to 24-48 hour jobs - Shared resources impact reproducibility - Cloud provides: - On-demand access for deadlines - Latest GPU architectures (V100, A100) - Scalability for large experiments

Cost-Benefit Analysis: - Purchasing 4× V100 GPUs: $20,000-30,000 - Power, cooling, maintenance: $2,000/year - Cloud: Pay only for usage, no maintenance - Break-even: >2 years of continuous use

1.13.3 11.3 Travel Justification

Why 4 Conferences? ($10,000) - Academic career requires conference presentations - Networking critical for: - Postdoc/faculty positions - Industry collaborations - Staying current with field - Student rate travel grants reduce out-of-pocket costs

1.13.4 11.4 Alternative Budget Scenarios

Minimum Budget ($35,000): - Reduce testbed ($10K): 3 switches, 5 edge nodes, 20 IoT devices - Cloud ($10K): Use only spot instances, apply for all credits - Travel ($8K): 3 conferences instead of 4 - Publications ($2K): Only essential open access

Maximum Budget ($70,000): - Enhanced testbed ($30K): 10 switches, 20 edge nodes, 100 IoT devices - Cloud ($25K): More GPU hours for extensive hyperparameter tuning - Travel ($15K): 5-6 conferences, international workshops

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1.14 Conclusion

The $53,000 budget is realistic and justified for a 4-year PhD program in AI-based network security. The major expenses (testbed equipment and cloud GPU) are essential for validating the proposed distributed IDS framework in real-world scenarios. Multiple funding sources (department grants, fellowships, cloud credits, industry partnerships) will be pursued to cover the budget.

Budget Highlights: - □ **Justified:** Every expense tied to research objectives - □ **Realistic:** Based on current market prices - □ **Flexible:** Minimum and maximum scenarios provided - □ **Fundable:** Multiple funding sources identified

Cost Optimization Strategies: - Apply for cloud research credits (potential $10K savings) - Seek industry equipment donations - Use conference travel grants - Apply for competitive fellowships (NSF GRFP, industry)

Return on Investment: - **Academic:** 5-10 publications, PhD degree - **Practical:** Open-source framework benefiting community - **Career:** Strong foundation for postdoc/faculty/industry positions - **Impact:** Improved security for billions of IoT devices and 5G networks

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Next: Refer to `collaboration.md` for potential funding and equipment partnerships, and `tools_and_frameworks.md` for software requirements.